

Dipartimento di Ingegneria e Scienza dell'Informazione

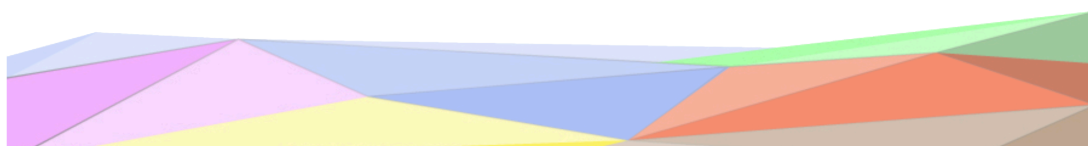
- KnowDive Group -

Schools and Kindergartens of Trento

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Francesco Magalini, December 2025

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1. Introduction

This document provides a comprehensive overview of the development process for the KGE project. It outlines the key stages, technical decisions, and implementation details that shaped the project from initial concept to completion. The code and this documents itself can be found on GitHub at the following [link](#)

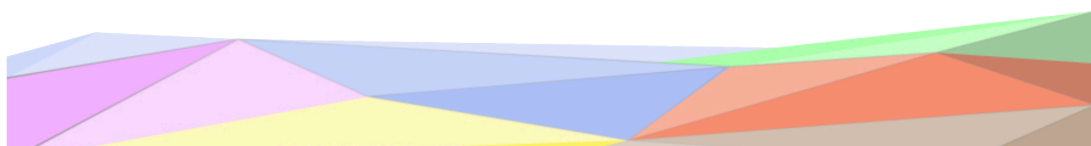
2. Purpose

2.1. Purpose Description

The project originated from a simple but rather useful informal idea: “to create a service that helps users find schools in the Trentino region by municipality and geographical position and explore details such as school type, number of classes and students and graduation rates.”

From this objective, the project’s **Domain of Interest** can be defined as all information related to educational institutions in Trentino and the municipalities of the Autonomous Province of Trento. This includes data about each institution’s facilities, the number of students and classes, and the graduation rate. The facilities included in this study are kindergartens, elementary and secondary schools (universities have been left out). The information related to these facilities as well as all the municipalities belonging to the autonomous region of Trento is part of the project domain. Moreover additional information such as the number of students and classes and graduation rates from each school is included.

Ultimately, the goal is to build a knowledge graph that serves as a unified repository of educational information for Trentino. This system should make it easier for both residents and visitors to



discover relevant insights about the region's educational opportunities. The following sections further detail the project's purpose through Personas and Scenarios.

2.2. Personas and Scenarios

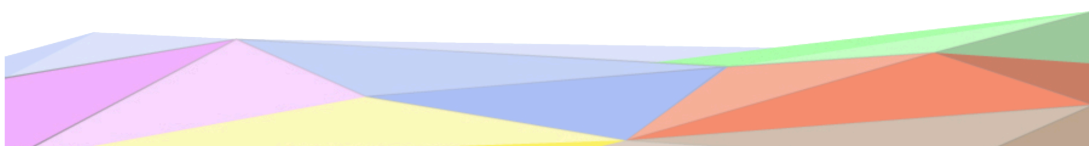
At the start of this section, we highlighted how essential persona analysis is for grasping the project's target domain and clarifying its goals. Personas serve as abstractions for real users: they capture their motivations, goals, and defining characteristics. When they accurately reflect real-world behaviors, personas function as a reliable reference point for guiding both design and development choices. In this work, the personas represent the types of users who will eventually interact with the output of the knowledge-graph engineering process. This subsection therefore outlines several user categories, each engaging with the system for their own reasons. Although all users may share the basic task of searching for schools in the Trentino region, their individual objectives can vary, which leads them to use the system, and interpret its results, in different ways.

- **Annamaria Verdi** is a mother of a child who is going to attend elementary school soon. She currently resides in Rovereto and she doesn't have a car, so she would like to look up all the educational facilities in her vicinity. She also would like to look up the school website to peek at the photos, read the educational offer and ensure her child has everything he needs.
- **Fabrizio Alighieri** is a 14 year old teenager who needs to decide which high school ("Liceo") is the best for him. He wants to focus his search on all the private schools near the city of Trento. He is also interested in contacting the school's secretary office to see if it is possible to visit the school in person during the open days.
- **Luciano Accorti** is a retired high school professor. He would like to launch a small private institute to make education more accessible for his municipality. Luciano needs a way to check all the schools in the vicinity, the type (public / private, primary / secondary) and how many students they have to see if his idea is viable and to carefully plan all the details.
- **Viviana Rossi** is an athlete, she has won the gold medal at the last winter olympics. She is a regular guest at the Sport Festival of Trento and now she wants to speak in high schools around Trentino. Viviana needs a handy way to make a tour, selecting all the schools with enough students.
- **Giuseppe Veronesi** is a public officer. He needs to organize the new math competition for the Trento province. For this task he needs to divide the territory into smaller regions in order to define the selection phases. The regions must be balanced by number of schools and students.

2.3. Competency Questions

Once the personas and their needs have been defined it can be more clearer how the project is intended to be used. We came up with a series of realistic Competency Questions based on the possible needs of the user base. These questions will be used to test the reliability and exhaustiveness of the final knowledge graph implementation.

1. Give me the complete list of schools from the Rovereto municipality.
2. Give me the number of high schools in San Michele All'Adige.
3. Give me the list of kindergartens from Lavis.
4. Give me the list of primary schools from Trento and nearby municipalities.
5. Give me the schools in the municipalities with the highest altitude



6. Give me the schools from the least populated municipalities
7. Give me the list of schools with the highest admission rates to the next year
8. Give me the list of private high schools from Pergine Valsugana, ordered by graduation rates.
9. Give me the schools with the lowest number of students.
10. Give me the schools with the highest number of students.
11. Give me the list of elementary schools within a 20km range from Trento

3. Information Gathering

This section describes, at top level, the work behind the information gathering phase.

3.1. Data Sources

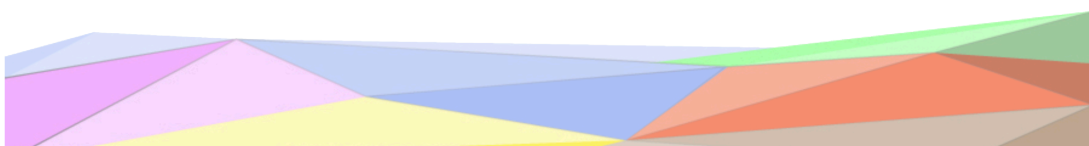
Three main sources of raw data have been selected in order to build the final knowledge graph.

3.1.1. Wikipedia

In order to obtain all the municipalities from the Trentino region, wikipedia was selected as a source. Starting from [this page](#), using **Python** and the **BeautifulSoup** package, the complete list was scraped from a table using attribute selectors and regexes. The municipalities from the Bolzano province that were present as well in the table have been left out. With this initial gathering, information such as name, population, density and altitude have been collected. Moreover for every entry of the table the link for the municipality own page has been collected. From this each single page have been scraped in order to obtain additional data such as postal code, istat code, latitude, longitude and bordering municipalities. Only the bordering municipalities that are municipalities of Trento (that is only the ones included in the dataset) have been considered. For the cleaning processes the **Polars** library was used for manipulating the csv file with high performance and ease of use.

166 rows

- **municipality** name of the municipality
- **province** name of the province
- **population** pop. from ISTAT 01/01/2024
- **surface** in km²
- **density** in ab/km²
- **altitude** in mslm
- **wikipedia link** link of the wikipedia page
- **postal code**
- **istat code**
- **sito**
- **bordering municipalities**
- **latitude** in the format: 46°09'N
- **longitude** in the format: 11°12'E
- **latitude** numeric
- **longitude** numeric



3.1.2. Vivoscuola

Vivoscuola is the Trentino school website with the goal of supporting the ongoing improvement of the Trentino education system and the contact between schools and families. This website contains of all the schools from kindergarten to secondary schools (universities are not included) in the Trento province. A .csv file was easily available and downloadable right from the website. It required a little bit cleaning, that was done with **Python** and **Polars**, and it contained almost all of the data properties that will be listed below.

Now in order to access the “Sistema informativo della scuola trentina” APIs we needed an unique school id that could be found on the Vivoscuola school page, inside the HTML code. Using **Python** and **BeautifulSoup** we first scraped the search page, after artificially changing the query parameters in order to obtain the full list of institutes and schools. With this process we gathered all the school page links. Now we scraped each page in order to get the API id, in order to speed up the process we used a **multithreaded** approach.

1012 rows

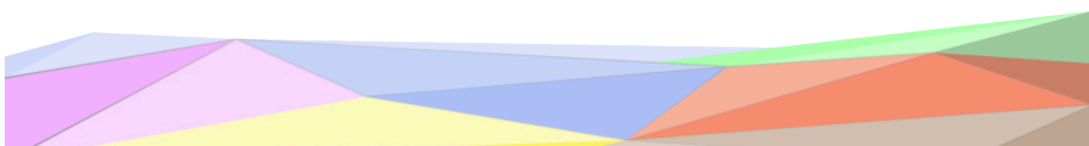
- **main institute** name of the institute
- **school** name of the school (empty if institute)
- **institute type**
- **administration type**
- **manager**
- **director**
- **educational coordinator**
- **address**
- **municipality** name of the municipality
- **telephone**
- **fax**
- **institute email**
- **management email**
- **secretary's office email**
- **website**
- **miur code**
- **vivoscuola institute link**
- **vivoscuola school link**
- **entity**
- **school type**

3.1.3. Sistema informativo della scuola trentina

Sistema informativo della scuola trentina is a digital service provider for schools in the Trentino region. It provides a digital registry for the professor as well as other tools for the management in order to facilitate the supervision of the necessary operations. This system provides open API endpoints that can provide data regarding the number of classes and students attending each school.

After obtaining all the unique school ids from the vivoscuola website we are able to query the APIs. Please note that the school link has been noted down in order to link this data with the vivoscuola dataset later if necessary. After some brief data cleaning the result is an array where every object is shaped like the following example:

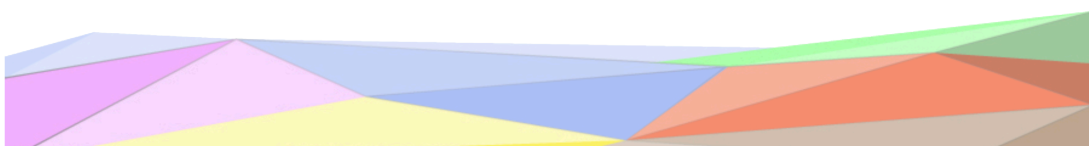
```
{  
  "school link": "https://www.vivoscuola.it/content/view/full/29266",  
  "api id": "0222059697",  
}
```



```

"grades": {
  "percentage admissions total": 75.24557956777997,
  "results": [
    {
      "percentages": [
        61.93181818181818,
        76.08695652173914,
        76.36363636363637,
        85.71428571428571
      ],
      "name": "Amessi 2022/23"
    },
    {
      "percentages": [
        61.76470588235294,
        67.83216783216784,
        91.80327868852459,
        95.0
      ],
      "name": "Amessi 2023/24"
    },
    {
      "percentages": [
        64.8936170212766,
        79.5774647887324,
        79.81651376146789,
        87.14285714285714
      ],
      "name": "Amessi 2024/25"
    },
    {
      "percentages": [
        78.43713278495888,
        84.78701825557809,
        86.82114409847937,
        90.57043073341094
      ],
      "name": "Amessi 2024/25 PAT"
    }
  ]
},
"students": {
  "current_students": [
    {
      "academic year": "2025/26",
      "school year": 1,
      "number of students": 173,
      "number of classes": 8
    },
    {
      "academic year": "2025/26",
      "school year": 2,
      "number of students": 159,
      "number of classes": 9
    }
  ]
}

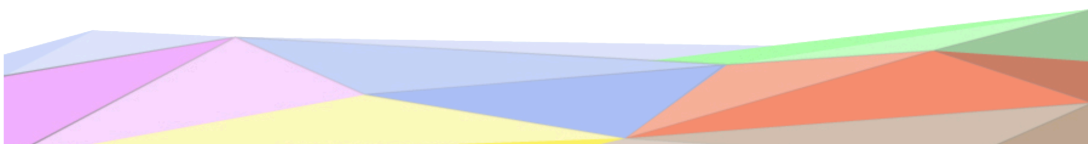
```



```

        "academic year": "2025/26",
        "school year": 3,
        "number of students": 134,
        "number of classes": 9
    },
    {
        "academic year": "2025/26",
        "school year": 4,
        "number of students": 62,
        "number of classes": 6
    }
],
"past_year_students": [
    {
        "academic year": "2024/25",
        "school year": 1,
        "number of students": 210,
        "number of classes": 9
    },
    {
        "academic year": "2024/25",
        "school year": 2,
        "number of students": 162,
        "number of classes": 9
    },
    {
        "academic year": "2024/25",
        "school year": 3,
        "number of students": 115,
        "number of classes": 7
    },
    {
        "academic year": "2024/25",
        "school year": 4,
        "number of students": 76,
        "number of classes": 6
    }
],
"past_2years_students": [
    {
        "academic year": "2023/24",
        "school year": 1,
        "number of students": 176,
        "number of classes": 8
    },
    {
        "academic year": "2023/24",
        "school year": 2,
        "number of students": 158,
        "number of classes": 9
    },
    {
        "academic year": "2023/24",
        "school year": 3,
        "number of students": 130,
        "number of classes": 9
    }
]

```



```

    },
    {
        "academic year": "2023/24",
        "school year": 4,
        "number of students": 64,
        "number of classes": 4
    },
    {
        "academic year": "2023/24",
        "school year": 5,
        "number of students": 23,
        "number of classes": 1
    }
]
}
}

```

4. Informal Modeling

4.1. Data Alignment

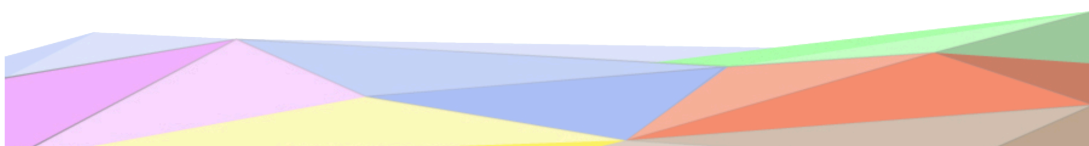
Data requires a final cleaning iteration in order to align the scraped content to the ER model presented below. For instance while parsing schools and municipalities both the Italian and German name were presented as one, or apostrophes and accents were often mixed. Using **polars** and **python** these errors were fixed and all the nomenclature was unified.

Finally five cleaned datasets in .csv format were generated as described below

- **Municipality**

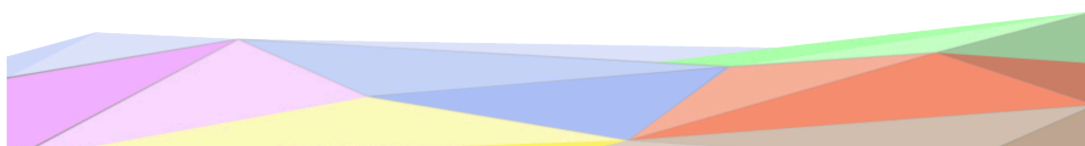
A complete list of municipalities, more precisely the “Comune” italian administrative region

- **municipality** name of the municipality
- **province** name of the city where the municipality belongs
- **population** number of people living in the municipality according to ISTAT at the 1st of January 2024
- **surface** number of square kilometers of surface
- **density** average number of people for every square kilometers
- **altitude** number of meters above the sea level
- **wikipedia link** wikipedia page of the municipality
- **postal code** italian postal code of the municipality
- **istat code** istat code of the municipality
- **website** website of the municipality (if available)
- **bordering municipalities** bordering municipalities (only those that are in this dataset)
- **latitude** latitude (*float*)
- **longitude** longitude (*float*)
- **Institute**



Institutes (“Istituti comprensivi”) are administrative units that bring together one or more schools in order to ensure educational continuity within the same cycle of education. Each institute is managed and administered by a single head teacher. Comprehensive schools were introduced in 1994 as an alternative to the organization of schools in mountain areas.

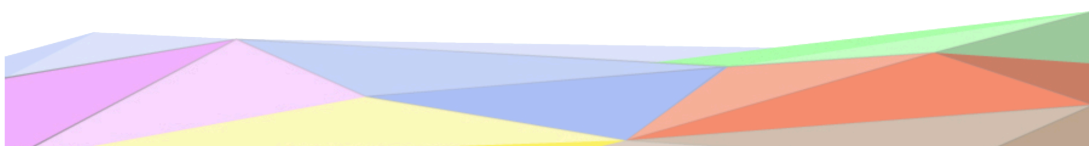
- **main institute** name of the institute
- **vivoscuola institute link** link of the institute on the vivoscuola platform
- **institute type** text describing the purpose of the institute (“istituzione scolastica e formativa” / “scuola dell’infanzia” / “circolo di coordinamento pedagogico”)
- **administration type** type of administration (“paritaria/equiparata”, “a carattere statale/provinciale”)
- **manager** name of the manager
- **director** name of the director
- **educational coordinator** name of the educational coordinator
- **address** text indicating the address
- **municipality** name of the municipality
- **telephone**
- **fax**
- **institute email**
- **management email**
- **secretary’s office email**
- **website** institute website
- **miur code** <https://www.mim.gov.it/glossario-iscrizioni-online>
- **wikipedia municipality link** link of the municipality on wikipedia
- **School**
- **school** name of the school
- **vivoscuola institute link** link of the institute on the vivoscuola platform
- **vivoscuola school link** link of the school on the vivoscuola platform
- **address** text indicating the address
- **municipality** name of the municipality
- **telephone**
- **institute email**
- **miur code**
- **admission rate** average percentage of students that are allowed to proceed to the next school year
- **wikipedia municipality link** link of the municipality on wikipedia
- **school type** (“infanzia”, “primaria”, “secondaria di primo grado”, “secondaria di secondo grado”)



- **Grades Statistics**
- **vivoscuola school link**
- **academic year** example: 2023/2024
- **school year** (*int*)
- **admission rate** average percentage of students that are allowed to proceed to the next school year
- **Students Information**
- **vivoscuola school link**
- **academic year** example: 2023/2024
- **school year** (*int*)
- **number of students**
- **number of classes**

4.2. ER Modeling

Developing an ER Model can be a useful abstraction for gathering ideas and improving the quality of later formal modeling. We can determine the necessary **Etypes**, and their **object attributes** as well as **data properties** based on our Competency Questions and the acquired datasets. From Figure 1 we can clearly see the Etypes: Institute, School, Municipality, Grade Statistics, Student Information, their attributes and also the relations between them.



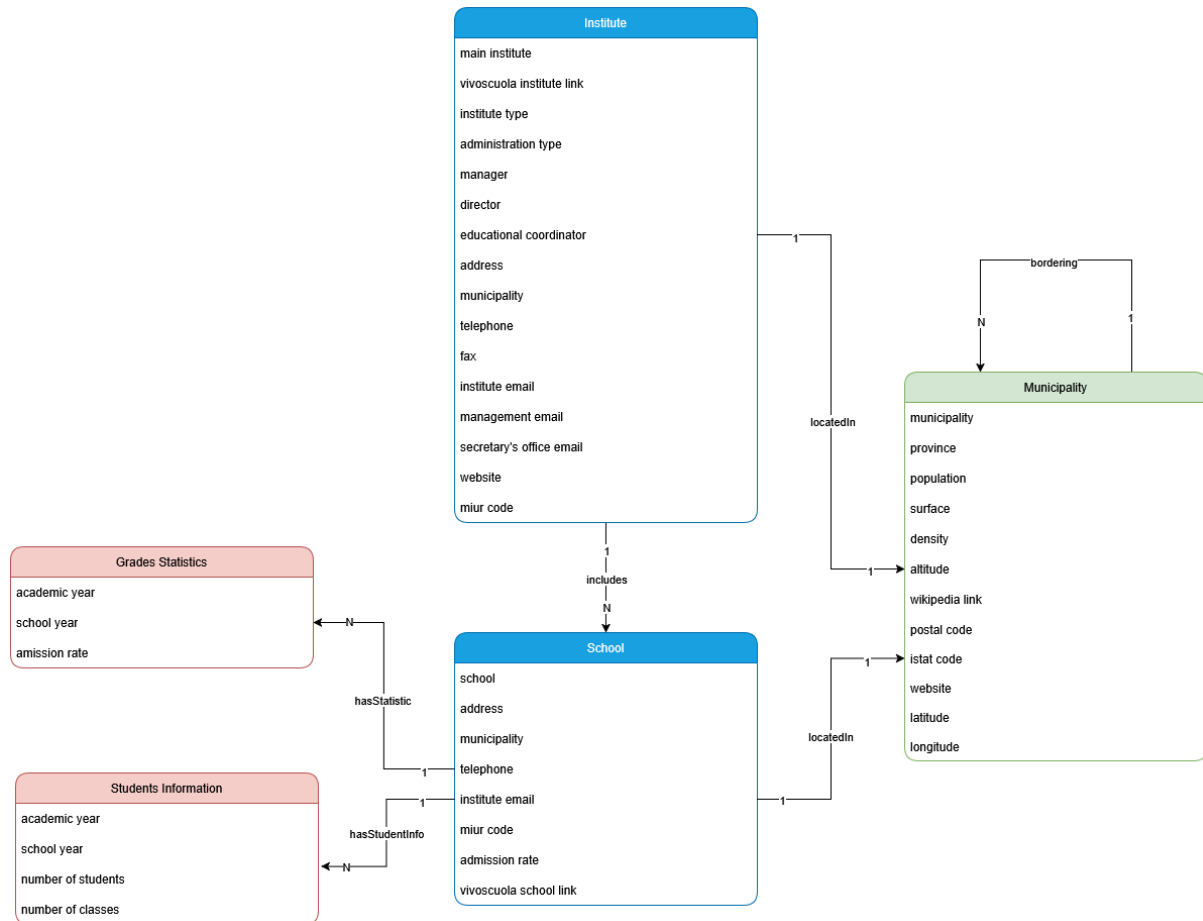


Figure 1: Representation of the ER model for the informal modeling phase

5. Formal Modeling

5.1. Language Definition

The language definition phase is important cause it ensures clarity and consistency by addressing ambiguity and diversity in natural and domain-specific languages. It enables accurate data annotation, minimizes misunderstandings, and supports seamless integration across systems. This fundamental step is crucial for effective communication and interoperability in complex, multilingual, or multi-domain projects.

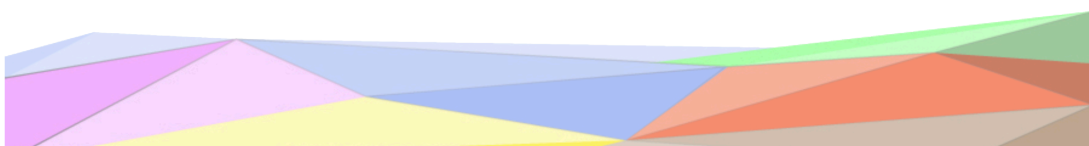
For this reason we relied on wiktionary and wikipedia whenever possible, by providing a clear multi lingual language definition for the most relevant terms in order to avoid any ambiguity

The outcomes produced during this phase can be found in the Github repository.

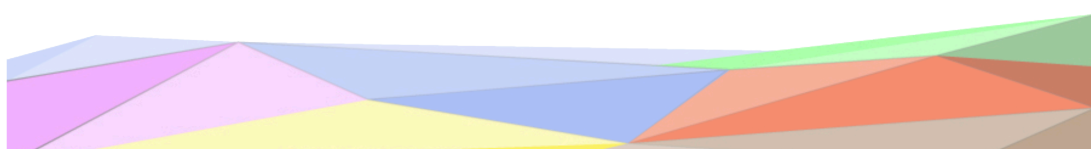
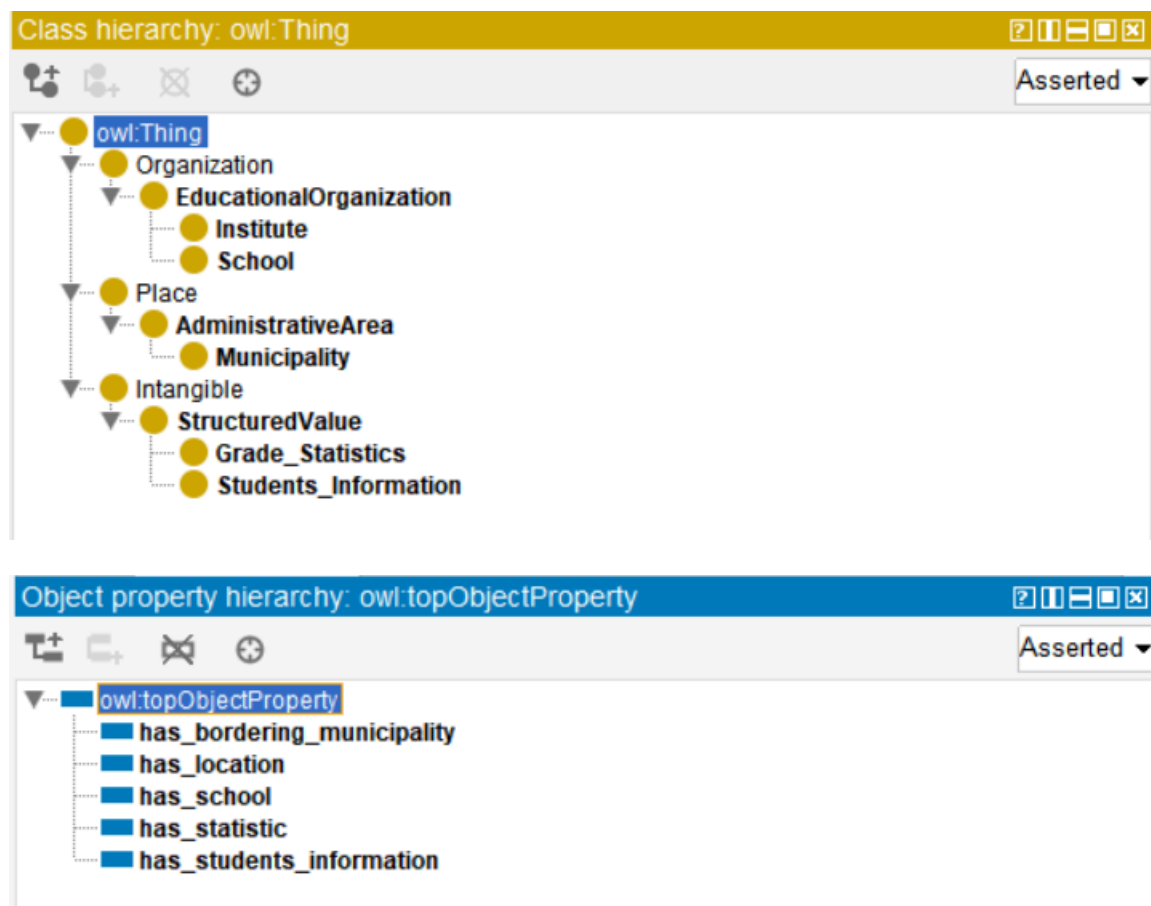
5.2. Knowledge and Entity Definition

The **Knowledge Definition** phase is a critical step within the iTelos methodology, aimed at formalizing and aligning the gained information to construct a Knowledge Graph. This phase consists in creating a unified representation of knowledge by defining teleontologies and aligning datasets. This report outlines the activities, decisions, and outcomes achieved in this phase.

We used **Protégé**, a widely recognized ontology editor, which facilitated the creation and management of the ontology by providing a comprehensive interface to define entity types, object



properties, and data properties. We also took inspiration from already existing entities from *schema.org*, however because these entities came with already predefined properties it was not possible to exactly map every single one of them according to our needs.



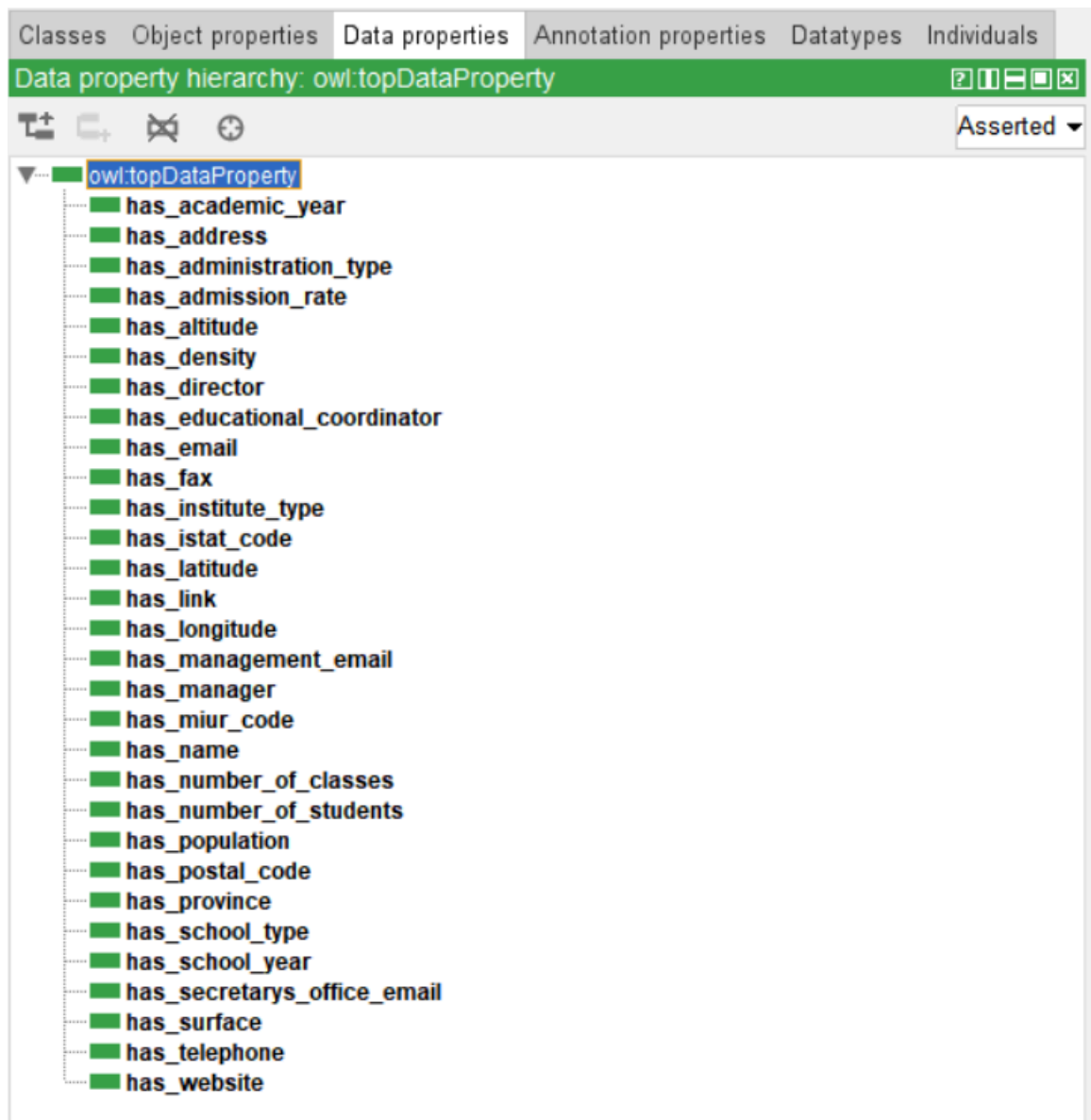
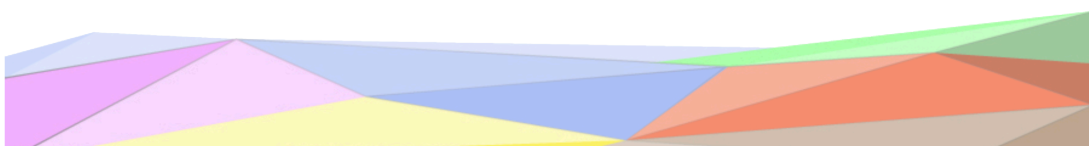


Figure 2: Examples of the schema definition from Protégé

The **Entity Definition** phase is the final step in the iTelos methodology, merging the knowledge and data layers into a unified Knowledge Graph. The input for this phase consists of cleaned and aligned data resources, as well as the teleontology created in previous stages. The objective is ensuring that entities across datasets are uniquely identified, matched, and mapped. The output of this phase includes the finalized KG and a set of mapping models in Turtle *.ttl* format.

With the aid of **Ontotext Refine** we were able to efficiently implement the desired mapping with a very handy user interface and the GREL scripting language. Once the mapping has been obtained it was fairly easy to produce Turtle files and ingest them with the **GraphDB** engine.



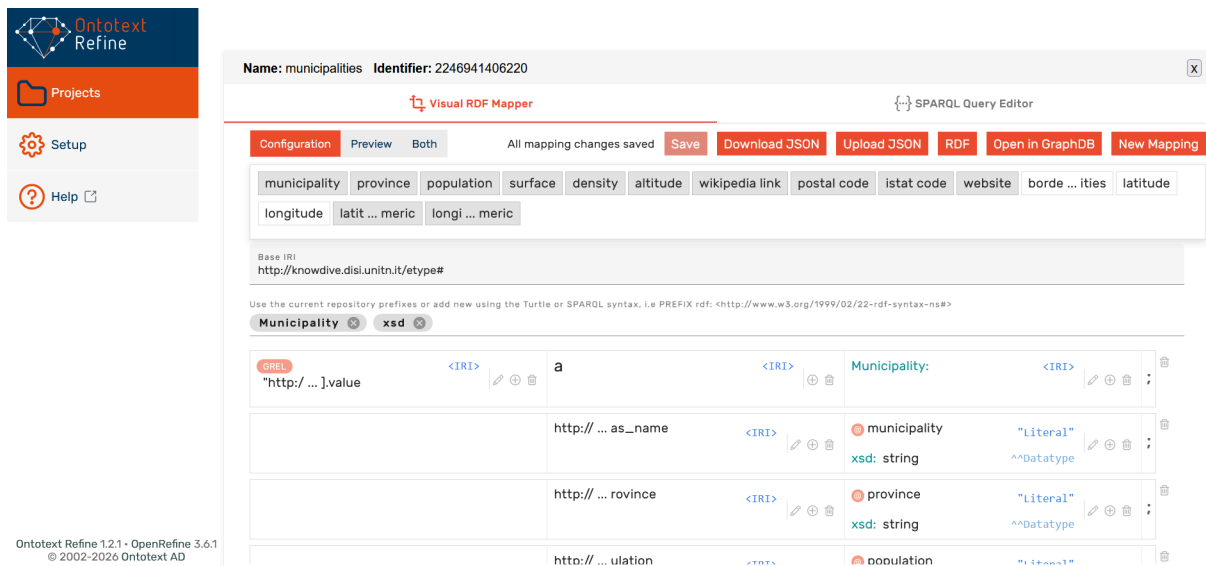


Figure 3: Examples of the mapping workflow from Ontotext Refine

6. Results

Finally we have successfully built our final knowledge graph, with a total of 166 municipalities and 1012 institutes and schools, including the relation between these entities and additional informations and statistics. Thanks to **GraphDB** we can fully explore it by using the **SPARQL** query language.

Below we've picked some examples from our queries, based on our **Competency Questions**, the full list can be found on GitHub.

Example 1 The following query (*cq n.3*) retrieves all the kindergartens from the municipality of "Lavis"

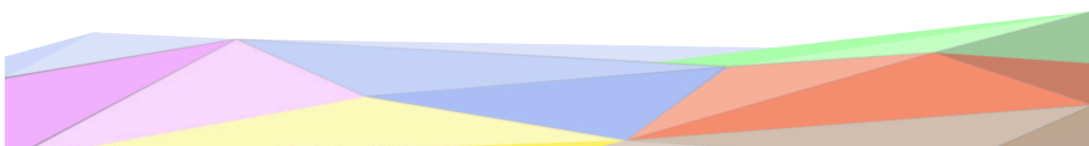
```
BASE <http://knowdive.disi.unitn.it/etype#>
PREFIX df: <http://knowdive.disi.unitn.it/etype#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

select ?s where {
  ?m rdf:type df:Municipality ;
  df:has_name "lavis" .
  ?s rdf:type df:School;
  df:has_location ?m .
  ?s df:has_school_type "infanzia"
}
```

Example 2 The following query (*cq n.6*) retrieves the top 20 schools located in the least populated municipalities

```
BASE <http://knowdive.disi.unitn.it/etype#>
PREFIX df: <http://knowdive.disi.unitn.it/etype#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

select ?s ?population where {
  ?m rdf:type df:Municipality ;
  df:has_population ?population .
  ?s rdf:type df:School;
```



```

    df:has_location ?m
} ORDER BY ASC(?population) LIMIT 20

```

Example 3 The following query (cq n.6) retrieves all the elementary schools located in a municipality within a 20km range from “Trento”.

The equirectangular approximation has been used for calculating the distance between two points. Although this formula is fairly simple it may not be the best regarding accuracy.

```

BASE <http://knowdive.disi.unitn.it/etype#>
PREFIX df: <http://knowdive.disi.unitn.it/etype#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX ofn: <http://www.ontotext.com/sparql/functions/>

select distinct ?s ?distance_km_approx where {
    ?m rdf:type df:Municipality;
    df:has_latitude ?lat0;
    df:has_longitude ?lon0;
    df:has_name "trento" .
    ?m1 rdf:type df:Municipality;
    df:has_latitude ?lat;
    df:has_longitude ?lon;
    BIND( (?lon - ?lon0) * ofn:cos( ((?lat + ?lat0)/2) * ofn:pi() / 180 ) AS ?dx ) .
    BIND( ?lat - ?lat0 AS ?dy ) .
    BIND(ofn:sqrt( (?dx * ?dx) + (?dy * ?dy) ) * 111.32 AS ?distance_km_approx) .
    ?s rdf:type df:School;
    df:has_school_type "primaria";
    df:has_location ?m1 .
    FILTER( ?distance_km_approx <= 20 )
} ORDER BY ASC(?distance_km_approx)

```

7. Conclusion

With this project we successfully learned how to implement a knowledge graph from scratch by focusing on every single step, from scraping and data collection to implementing queries. Learning how to use popular tools like **Protégé**, **Ontotext Refine** and **GraphDB** was undoubtedly a positive experience. While the web scraping phase presented its own set of challenges, like dealing with inconsistent data and align the raw information to the desired ontologies, it has been a useful lesson.

Indeed there is definitely room from improvement, additional data sources can be brought in, such as data from universities, professors, courses, etc. Moreover the geolocation capabilities could be enhanced by attaching transportation data or precise latitude and longitude information from services like Google Maps or OpenStreetMaps.

Nonetheless we showed how from a limited number of data sources, a compact but valuable knowledge graph representation was possible.

